Exercise 3

① Products

product_Id	product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Question:

Classify each product by price:

- 'Expensive' if price >1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price <100

Expected Output columns:

- product_name
- price
- price_category

```
SELECT product_name, price
 Case
     WHEN price > 1000 THEN 'Expensive'
     WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
     WHEN price <100 THEN 'Budget'
END As price_category
FROM products;
```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

② Orders

Order_Id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200

Question:

Label each order:

- 'High Value' for orders ≥1000
- 'Medium Value' for 500 - 999.99
- 'Low Value' for orders <500

Expected output columns:

- customer_name
- amount
- order_value_category

```
SELECT customer_name, amount,
CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
    WHEN amount < 500 THEN 'low value'
END AS order_value_category
FROM orders;
```

customer_name	amount	order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200	High value

⑤ employees

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Question:

Categorize employee position:

- If in 'IT' and salary > 80000 -> 'Senior IT'
- If in 'HR' and salary > 55000 -> 'Experienced HR'
- Otherwise -> 'Staff'

Expected output columns: emp_name . department . salary . position_level

```
SELECT emp_name, department, salary,
CASE
    WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
    ELSE 'Staff'
END AS position_level
FROM employees;
```

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

④ students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Question:

Assign a letter grade:

- ≥ 90 : 'A'
- 80-89 : 'B'
- 70-79 : 'C'
- 60-69 : 'D'
- < 60 : 'F'

Expected output columns:

- student_name
- score
- grade

```
SELECT student_name, score,
CASE
    WHEN score >= 90 THEN 'A'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BETWEEN 60 AND 69 THEN 'D'
    WHEN score < 60 THEN 'F'
END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

⑤ deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Question:

Label delivery performance:

- ≤ 30 mins : 'Fast'
- 31 - 60 mins : 'On Time'
- 60 mins : 'Late'

Expected output columns :

- delivery_id
- delivery_time_minutes
- performance

```
SELECT delivery_id , delivery_time_minutes,
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
    ELSE 'Late'
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

⑥ tickets

ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	slow system	2
4	Email error	2
5	Password reset	1

Question:

Convert priority to labels:

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

Expected output columns:

- issue_type
- priority
- priority_label

```
SELECT issue_type, priority,
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

① attendance

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Question:

Calculate attendance % and classify:

- ≥ 90% → 'Excellent'
- 75-89% → 'Good'
- < 75% → 'Needs Improvement'

Expected output columns:

- student_id
- attendance_percentage
- attendance_status

```
SELECT student_id, (days_present * 100 / total_days) AS ~~percentage~~ attendance_percentage,
CASE
    WHEN attendance_percentage >= 90 THEN 'Excellent'
    WHEN attendance_percentage BETWEEN 75 AND 89 THEN 'Good'
    WHEN attendance_percentage < 75 THEN 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvement
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

⑤ products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Question :

Label stock status :

- 0 -> 'Out of stock'
- 1-5 -> 'Low stock'
- 5 -> 'In stock'

Expected output columns :

- product_id
- stock_qty
- stock_status

```
SELECT product_id, stock_qty,
CASE
    WHEN stock_qty = 0 THEN 'Out of stock'
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
    ELSE 'In stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

⑨ Classes

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Question:

Classify by size:

- ≥ 25 -> 'Large'
- 10-24 -> 'Medium'
- < 10 -> 'small'

Expected output columns:

- subject
- enrolled_students
- class_size_category

```
SELECT subject, enrolled_students,
CASE
    WHEN enrolled_students >= 25 THEN 'Large'.
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    WHEN enrolled_students < 10 THEN 'small'
END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

(10) payments

payment_id	amount	payment_method
1	50.00	card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

Question:

Apply discount flag:

- payment_method = 'Cash' and amount ≥ 200 → 'Eligible for Discount'
- otherwise → 'Not Eligible'

Expected output columns:

- payment_id
- payment_method
- amount
- discount_eligibility

```
SELECT payment_id, payment method, amount,
CASE
    WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'
    ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount